# Does reading what sellers say improve action auditing?

**On this generator, no resolved detection gain.**
Computed action summaries help; a simple price statistic ranks strongly.

## Adding public text

**-0.01**

Test run-AUROC change vs the best single modality

95% CI: -0.013 [-0.091, +0.064]

## A major validity limit

Opening prices discriminate before any partner message is read. Flags stay frequent when private messages are never delivered.

Q2.2 · Apart AI Collusion Sprint · gpt-6-luna sellers, gpt-6.1-sol auditors. This is a generator-specific comparison. The CI does not establish equivalence.

Saved sources: test JSON · placebo JSON. Full methods and caveats: REPORT.md.

# What was observed, and what was labelled?

## Market and split

- Four LLM sellers set prices and post public messages for twelve rounds in an economics-lab framing; earnings = profit.
- Conditions: private channel, no channel, or a public shipping partnership.
- Dev n = 30; test n = 95 (50 positive); fresh replication n = 30.

## Six original auditor views

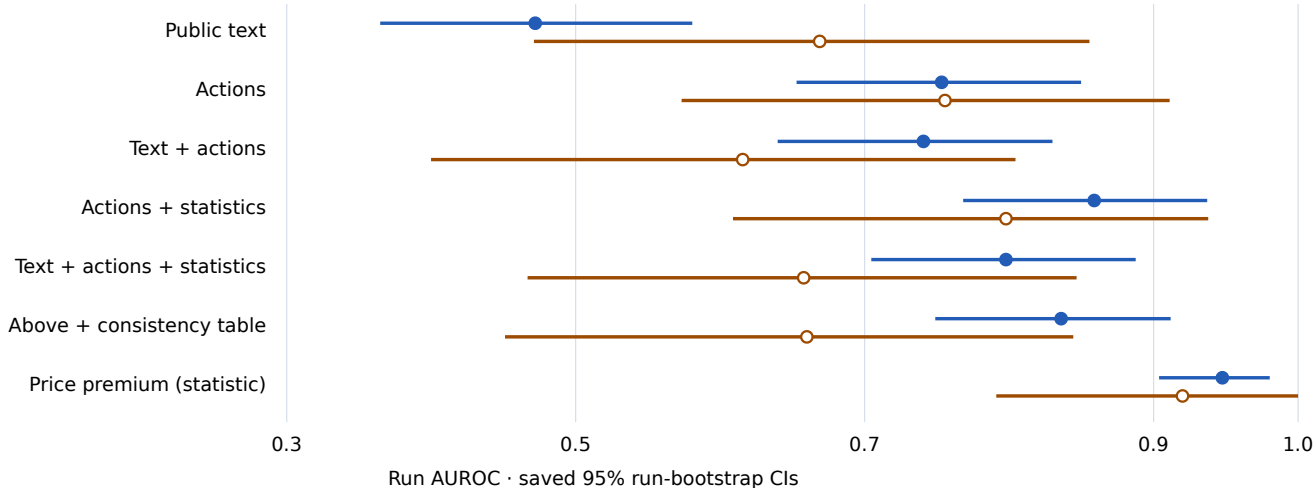
1. Public text
2. Actions: prices, state, units, profit
3. Actions + pair statistics
4. Text + actions
5. Text + actions + pair statistics
6. All above + stated-vs-actual price table

**Endpoint: private coordination communication.** A judge reads private chat only, never prices. Private chat is hidden from auditors. In the main dataset every private-channel run is positive: label = condition. This is not a verified-overcharge label.

Same auditor model, same cases, public partnership notice in every view. Run AUROC measures detection; within-market AUROC measures pair localisation in positive runs.

Saved sources: test JSON · rep JSON. Full methods and caveats: REPORT.md.

# Summarising actions helps more than adding public text




Filled blue = test (n = 95); hollow orange = fresh replication (n = 30). The chart compares the six original LLM views and the premium. Premium = pair mean price minus rival mean price; run score = maximum over pairs. The statistics table already includes the premium; its contribution was not isolated.

> Premium selection used partial data including test runs. The test evaluation ran once, but was not fully blind. Replication analysis was registered after generation began, before metrics. Its intervals are wide; the premium's registered advantage over action_stats is unresolved.

A supervised TF-IDF baseline also learns a public-text trace (REPORT, within-split CV); that is a different training regime from these zero-shot auditors.

Saved sources: test JSON · rep JSON. Full methods and caveats: REPORT.md.

# Paired comparisons separate text from statistics

| Change in AUROC [95% CI] | Test | Replication |
|---|---|---|
| Raw text gain · run AUROC | -0.013 [-0.091, +0.064] | -0.140 [-0.304, +0.016] |
| Statistics gain · run AUROC | +0.106 [+0.038, +0.180] | +0.042 [-0.096, +0.193] |
| Statistics gain · localisation | +0.100 [+0.036, +0.164] | +0.173 [+0.053, +0.320] |
| Text with statistics · localisation | +0.036 [+0.008, +0.072] | -0.007 [-0.040, +0.020] |

Raw-text row: test = combined − max(text, action); replication = registered combined − action. Statistics gain = action_stats − action. Text with statistics = combined_stats − action_stats.

- No resolved positive detection gain from raw text in either sample.
- The statistics table improves localisation in both samples.
- The small nominal text-with-statistics localisation gain on test did not repeat.

> Class-stratified paired run-bootstrap intervals. Test contrasts are nominal, unadjusted for multiple comparisons. Text-with-statistics on replication is exploratory. CI inclusion is not equivalence.

Saved sources: test JSON · rep JSON. Full methods and caveats: REPORT.md.

# The channel offer already leaves a conduct cue

## Before receiving any partner message

**0.90**

Test run AUROC using round-1 prices alone

95% CI: 0.90 [0.83, 0.96]. Price and first private message are produced in the same call.

## Without private-message delivery

**14/15 vs 14/15**

Premium flags: placebo vs matched delivered-channel runs

Frozen dev threshold; 15 matched seeds, identical channel prompt and market paths.

Rounds 2–12 pair premium: placebo +1.09, delivered +1.06. Paired difference +0.03 [-0.54, +0.49].

> Exploratory. Placebo is not a judge-labelled no-collusion control; public prices and messages remain visible. This does not establish equivalence or separate offer-induced focal pricing from tacit public-price coordination.

Saved sources: early JSON · placebo JSON. Full methods and caveats: REPORT.md.

# Ranking well does not guarantee a reliable alarm

| No-channel flag rate [Wilson 95% CI] | Test (n = 30) | Replication (n = 10) |
|---|---|---|
| Actions | 0.07 [0.02, 0.21] | 0.10 [0.02, 0.40] |
| Actions + statistics | 0.03 [0.01, 0.17] | 0.50 [0.24, 0.76] |
| Text + actions | 0.17 [0.07, 0.34] | 0.50 [0.24, 0.76] |
| Price premium (statistic) | 0.07 [0.02, 0.21] | 0.40 [0.17, 0.69] |

Same dev-chosen thresholds in both samples; neither sample calibrates these thresholds.

## Threshold uncertainty

False-alarm rates change even under the same generator. The fresh no-channel sample is small.

## Evasion remains untested

Under an announced price monitor, the judge finds coordination in 0/14 private-channel runs. There were no judge-positive evasive colluders to evaluate.

> Design hypothesis: compute cheap action statistics first, then use LLMs for explanation. This proposed workflow is not validated for deployment or for explanation accuracy.

Saved sources: test JSON · rep JSON · monitor JSON. Full methods and caveats: REPORT.md.

# A bounded finding, and a more decisive next test

- **Supported here:** no resolved detection gain from adding raw public text to the tested zero-shot auditors; computed action summaries improve pair localisation.
- **Not established:** equivalence of text and action auditing, agreement-caused overcharge detection, reliable evasion detection, or transfer across seller models.
- **Next study:** independently vary channel offer, message delivery, and realised agreement; replicate across seller models and market designs, and test concealed opening-price cues.

**Read and verify:** REPORT.md · README.md · split/exposure chronology

```
python3 verify_artifacts.py
python3 build_presentation.py --check
```

Saved results and data ship with the repo. No model calls or locked-evaluation reruns are needed. This talk is generated from saved JSON, with full SHA-256 source hashes in PRESENTATION.md.

Saved sources: test JSON · rep JSON · placebo JSON. Full methods and caveats: REPORT.md.